

Problem 1

- (a) Let A be the set of all functions $f : \{0, 1\} \rightarrow \mathbb{N}$

Let $g : A \rightarrow \mathbb{N} \times \mathbb{N}$ s.t. $g(f) = (f(0), f(1))$

Injective:

Let $f, h \in A$ s.t. $g(f) = g(h)$ thus this means that $(f(0), f(1)) = (h(0), h(1))$

This implies that $f = h$ since they map the same input to the same output for their whole domain thus g is injective.

Surjective:

Let $a, b \in \mathbb{N}$ thus consider $(a, b) \in \mathbb{N} \times \mathbb{N}$

Consider the function f that maps 0 to a and 1 to b . This function is in A . Thus, $g(f) = (a, b)$

Thus g is bijective, thus by proven in class since $\mathbb{N} \times \mathbb{N}$ and has a bijection to A we know that A is countable.

- (b) Consider F_n which is all functions from $\{1, \dots, n\}$ to $\mathbb{N}$.

Using a similar proof to (a) we can show that F_n has a bijection to $\mathbb{N}^n$, the Cartesian product of $\mathbb{N}$ taken n -times.

Shown in class that finite number of Cartesian products is finite thus each F_n is countable.

Then proven in class we showed that countable union of countable sets is countable, and since we only take $|\mathbb{N}|$ number of unions this is a countable number thus the set is countable.

- (c) Let A be the set of all functions $f : \mathbb{N} \rightarrow \{0, 1\}$

Let $g : A \rightarrow \{\text{infinite binary sequences}\}$ s.t. $g(f) = (f(1), f(2), \dots)$

Injective:

Let $f, h \in A$ s.t. $g(f) = g(h)$ thus this means that $(f(1), f(2), \dots) = (h(1), h(2), \dots)$ thus $f(1) = h(1), f(2) = h(2), \dots$ thus $f = h$ thus g is injective.

Surjective:

Let $s = (x_1, x_2, \dots)$ be a infinite binary sequence. Thus pick the function f in A s.t. $f(1) = x_1, f(2) = x_2, \dots$ thus $g(f) = s$

Thus g is bijective.

Now assume for contradiction that the set of infinite binary sequences is countable. Thus, we can list them $s_1, s_2, \dots$

Now define the sequence t s.t. $t(n) = 1$ if s_n at the n th term is 0 and 0 if the n th term is 1.

t cannot be in our list because if it were lets say it was s_m but at the m th spot we flipped it in t thus this is a contradiction.

Thus, A is not countable.

- (d) Let A be the set in the question.

Let $g : A \rightarrow \bigcup_{N \in \mathbb{N}} \{\text{binary sequences of length } N - 1\}$ s.t. for f s.t it is eventually zero at $N \in \mathbb{N}$

this means for all $n \geq N$, $f(n) = 0$ thus define g s.t. $g(f) = (f(1), f(2), \dots, f(N-1))$

Injective:

Let $f, h \in A$ s.t. $g(f) = g(h)$ thus we know that both f, h both become zero at the same spot lets call it N and that $f(1) = h(1), \dots, f(N-1) = h(N-1)$ thus these are the same functions since everything after and at N is the same. Thus, $f = h$ thus this function is injective.

Surjective:

Let $s \in \bigcup_{N \in \mathbb{N}} \{\text{binary sequences of length } N-1\}$. WLOG let s be length $N \in \mathbb{N}$. Thus pick the function in A s.t. $f(1) = x_1, f(2) = x_2, \dots, f(N) = x_N$ s.t. x_i are the i th term in s , and that f is zero after and at N

Thus $g(f) = s$

Thus A is bijective to the set of all finite binary sequences. Thus, for each size the number of sequences is 2^N where N is the size, thus each set is finite thus finite then we do a countable union on these finite sets thus this whole set is countable.

(e) Let A be the set in the question.

Let $g : A \rightarrow \bigcup_{N \in \mathbb{N}} \mathbb{N}^N$ s.t. for f s.t it is eventually constant at $N \in \mathbb{N}$ this means for all $n \geq N$, $f(n) = m$ for $m \in \mathbb{N}$ thus define g s.t. $g(f) = (f(1), f(2), \dots, f(N))$

Injective:

Let $f, h \in A$ s.t. $g(f) = g(h)$ thus we know that both f, h both become constant at the same spot lets call it N and that $f(1) = h(1), \dots, f(N) = h(N)$ thus these are the same functions since everything after and at N is the same. Thus, $f = h$ thus injective. Surjective:

Let $s = (x_1, x_2, \dots, x_N)$. WLOG s is length N . Thus pick the function in A s.t. $f(i) = x_i$ for $i \in \{1, \dots, N\}$, and f is constant at and after N . Thus, $g(f) = s$

Thus g is bijective. Thus, since finite cartesian products are countable we are doing countable unions on countable sets thus the whole set is countable.

(f) Let A be the set of all two-element subsets of $\mathbb{N}$. Thus $A = \{\{a, b\} \subset \mathbb{N} \mid a \neq b\}$

Let $g : A \rightarrow \{(a, b) \in \mathbb{N} \times \mathbb{N} \mid a < b\}$ s.t. $g(\{a, b\}) = (a, b)$ if $a < b$ or (b, a) if $b < a$.

Injective:

Let $c = \{a, b\}, d = \{e, f\} \in A$ s.t. $g(c) = g(d)$. WLOG let $a < b$ and $e < f$. Thus, since $g(c) = g(d)$ we know that $a = e$ and $b = f$. Thus, we know that c, d are the same subset.

Surjective:

Consider (a, b) consider the two element subset $\{a, b\}$ thus $g(\{a, b\}) = (a, b)$.

Thus, A has a bijection to a subset of a countable set $\mathbb{N} \times \mathbb{N}$ thus it is countable.

(g) Let A be all finite subsets. Do a similar proof in A , but do the bijection to $\bigcup_{N \in \mathbb{N}} \{(x_1, \dots, x_N) \in \mathbb{N}^N \mid x_1 < \dots < x_N\}$. Thus since we know that finite cartesian products are countable, and we do a countable union on them we know that the whole set is countable.

Problem 2

(a) Reflexivity: Since $f(x) = f(x)$ thus $x \sim x$

Symmetry: If $x \sim y$ then $f(x) = f(y)$ then $f(y) = f(x)$ thus $y \sim x$

Transitivity: If $x \sim y$ and $y \sim z$ then $f(x) = f(y)$ and $f(y) = f(z)$ thus $f(x) = f(z)$ thus $x \sim z$

Thus this is an equivalence relation.

Let $g : \{\text{equivalence classes of } A\} = C \rightarrow B$ s.t. $g([x]) = f(x)$

Injective:

Let $[a], [b] \in C$ s.t. $g([a]) = g([b])$. Thus, we know that $f(a) = f(b)$ thus we know that $a \sim b$ thus $[a] = [b]$ Thus injective.

Surjective:

Let $b \in B$ since f is surjective there exists an $a \in A$ s.t. $f(a) = b$ then take the equivalence class a is in thus $g([a]) = f(a) = b$ thus surjective.

Thus, g is bijective.

(b) Reflexivity: Since $A = A$ we can use the identity map from A to A which is a bijection thus we know that $|A| = |A|$ thus $A \sim A$.

Symmetry: If $A \sim B$ then $|A| = |B|$ thus $|B| = |A|$ thus $B \sim A$

Transitivity: If $A \sim B$ and $B \sim C$ then there is a bijection from A to B and B to C thus compose these bijections and we get a bijection from A to C thus $|A| = |C|$ thus $A \sim C$.

Problem 3

Problem 4

Let $f : (X^\omega)^\omega \rightarrow X^\omega$ s.t. given $S = (s_1, s_2, \dots)$ s.t. $s_i = (x_{i,1}, x_{i,2}, \dots)$ we can define $f(S) = (y_{\pi(1,1)} = x_{1,1}, y_{\pi(2,1)} = x_{2,1}, \dots)$ s.t. $\pi(i, j) = \frac{(i+j-2)(i+j-1)}{2} + j$ gives the order which is basically diagonal order.

Injective:

Let $a, b \in (X^\omega)^\omega$ s.t. $f(a) = f(b)$ thus we know that all items in the sequence $f(a), f(b)$ are the same. Thus, since the function f will map spot i, j to $\pi(i, j)$ we thus know that each i, j in a, b are the same thus $a = b$ thus they are injective.

Surjective:

Let $s \in X^\omega$ consider the element $a \in (X^\omega)^\omega$ s.t. the (i, j) th spot of a is equal to $x_{\pi(i, j)}$ in s . Thus, $f(a) = s$ thus surjective.

Thus we have found a bijection thus they have the same cardinality.

Problem 5

- (a) Let X be a well-ordered set. Let A be a bounded above subset of X . Thus consider the set $U = \{u \in X \mid u \geq a \text{ for all } a \in A\}$.

This is non empty since A is bounded above. Thus, since X is well-ordered any nonempty subset has a min thus we know found the least upper bound.

- (b) Let X be a well-ordered set. Let A be a nonempty subset of X . Let B be a nonempty subset of A . Thus, B is also a subset of X . Thus, B has a min. Thus, A is well-ordered.

Consider $\mathbb{R}$ and the subset $\mathbb{Q}$. Consider the subset of $\mathbb{Q}$ $A = \{x \in \mathbb{Q} \mid x < \sqrt{2}\}$. The supremum of this set is $\sqrt{2}$

- (c) Let $(a, b), (c, d) \in A \times B$ with $(a, c) \neq (c, d)$. Assume $(a, b) < (c, d)$. Assume for contradiction that $(c, d) < (a, b)$

Thus, we have that either $a < c$, or $a = c$ and $b < c$, and we have that either $c < a$, or $c = a$ and $c < b$. If $a < c$ then $a \neq c$ thus this implies $c < a$ thus this cannot hold. If $a = c$ and $b < d$ then this also implies $d < b$ which is a contradiction thus if $(a, b) < (c, d)$ then $(c, d) \not< (a, b)$

Let $(a, b) \in A \times B$ then Assume $(a, b) < (a, b)$ then we have that either $a < a$ or $a = a$ and $b < b$ well in both cases there are contradictions from the ordering of A and B thus $(a, b) \not< (a, b)$

Let $(a, b), (c, d), (e, f) \in A \times B$. Assume $(a, b) < (c, d)$ and $(c, d) < (e, f)$.

Thus, either $a < c$ or $a = c$ and $b < d$.

If $a < c$ then if $c < e$ then $a < c < e$ thus $(a, b) < (e, f)$. If $c = e$ then $a < c = e$ thus $a < e$ thus $(a, b) < (e, f)$

If $a = c$ and $b < d$ then if $c < e$ then $a = c < e$ thus $a < e$ thus $(a, b) < (e, f)$. If $c = e$ and $d < f$ then $b < d < f$ thus $b < f$ and $a = e$ thus $(a, b) < (e, f)$.

Thus, that is an order relation.

Assume A, B are well-ordered. Let C be a nonempty subset of $A \times B$

Let $C_A = \{a \in A \mid \exists b \in B \text{ s.t. } (a, b) \in C\}$

Thus since A is well-ordered we can find a min of C_A , and it is nonempty since C is nonempty.

Let α be the mins of C_A . Let $C_B = \{b \in B \mid (\alpha, b) \in C\}$. This is nonempty since C_A is nonempty and that α exists. Thus since B is well-ordered take the min of C_B call it β . Thus, consider (α, β)

For $(c, d) \in C$ s.t. $(c, d) \neq (\alpha, \beta)$ since α is the minimum A in C we know that α is always either less than or equal to c if it is equal then we are done. If they are equal then we know that $\beta \neq d$ since $(\alpha, \beta) \neq (c, d)$. Since β is the min B of all elements of the form (α, f) s.t. $f \in B$ we know that $\beta < d$ thus (α, β) is the min element. Thus, $A \times B$ is well-ordered.

- (d) Let X be a well-ordered set. Let $Y = P(X)$

Let the ordering of Y be for $A, B \in P(X)$ $A < B$ if and only if the least element in $(A \setminus B) \cup (B \setminus A)$

is in B and not A basically the first element where A and B differ determines the order.

Total order:

Let $A, B \in P(X)$ s.t. $A \neq B$. If they are the same then since we take out what they overlap in the set $(A \setminus B) \cup (B \setminus A)$ the least will only be in one set. Thus either $A < B$ or $B < A$

For A by itself we have that $(A \setminus B) \cup (B \setminus A)$ is empty, so it is undefined at this operation.

Let $C \in P(X)$. Assume $A < B$ and $B < C$.

$$(A \setminus c) \cup (C \setminus A) = ((A \setminus B) \cup (B \setminus A)) \cup (A \setminus c) \cup (C \setminus A)$$